Veer Technical Specification

Course: Computer Science Senior Design I
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Traces to: Requirements Document **version 1.0** (October 30, 2026)

1. Version and Change Log

Version	Date	Change	Author
D0 1.0	2026-10-02	High-level design: C1 to C4, E1 to E4, interfaces I1 to I9, decisions D01 to D10.	Team
D1 1.1	2026-10-08	C2 and C3 detailed: data model, allow-list validation, filter-then-rank selection, I2/I4 contracts, technology stack.	Team
D2 1.2	2026-10-13	C1 state machine, C4, and glue components. I1 split into I1a/I1b/I1c and approval replies moved to I2 (R-04): D0's I1 mixed unrelated jobs and could not be tested alone. UsageEvent fields renamed (provider → service, attemptedAt → occurredAt, outcome values ok/error/throttled) and estCost added. MuteError exit and network/GPS-loss edge cases (EC-33, EC-34) added after peer review.	Team
1.0	2026-10-30	Consolidated D0 to D2 into this specification, traced to Requirements Document v1.0. Two small changes since the Week 7 review: (1) auth, config, and usage endpoints now use the `/api/v1/` prefix, because D1 §5.4 requires it for every endpoint and D2 had left it off; (2) the I2 speech stream path is fixed as `/api/v1/trips/{tripId}/speech` so C1 and C2 can be built against it (audio encoding stays open under G-08). GitHub Actions selected for CI.	Aiden Ward

2. Architecture Summary

Veer is a client-server system with a layered backend and a pipeline for speech (D0 Block Diagram; full design in D0, D1, and D2).

Components. C1 Mobile Navigation is the Android app. It is the only component that can change the active route or control the microphone, and it runs the trip state machine (D2 Figure 1). The backend holds three logical modules: **C2 Speech Understanding** turns an utterance into a validated, allow-listed constraint; **C3 Route Planning** turns constraints into a route proposal by filtering then ranking candidates; **C4 Usage Monitor** counts and caps provider calls. Four glue components support them: authentication, notifications, configuration, and logging. External dependencies are **E1** Android, **E2** OpenAI, **E3** Google Maps Platform, and **E4** Open-Meteo.

Deployment. C1 installs on the driver's own phone. C2, C3, C4, and the glue components run as one Node.js/TypeScript service on Render with a managed PostgreSQL database. They are modules in one service, not microservices (D0 §6). Provider API keys exist only on the server.

Data flow (D0 Data Flow, D0 Usage Flow).

- *Speech to route.* C1 buffers an utterance in memory (I1a) and streams it to C2 (I2). C2 gets a transcript and candidate constraint from E2 (I3), checks it against the allow-list, and returns it. C1 sends the constraint with trip context to C3 (I4). C3 fetches routes and places from E3 (I5) and, for rain, weather from E4 (I6), then returns a proposal. C1 speaks a confirmation of 15 words or fewer, waits for "okay" (interpreted by C2 over I2), and activates the route within 3 seconds. Audio is discarded after each step.
- *Mute.* A tap or the local phrase "Veer, stop listening" closes the send gate, clears the buffer, cancels in-flight requests, and stops the mic on the phone. No network call is involved.
- *Usage.* Before each paid call C2 and C3 ask C4 for quota; after it they record a usage event (I7, I8). Developers read per-trip totals through I9. No audio, transcript, or location reaches C4.

3. Interfaces and Data Formats

3.1 Component Boundaries

Every network request carries `requestId`. Every failure returns `Error = {code: string, message: string, retryable: boolean}`, and no failure ever applies a route (D0 §4).

Boundary	Inputs	Outputs	Format	Protocol	Error behavior
I1a C1 ↔ E1 Mic	micControl (start, stop, forceStop); MuteEvent	PCMFrame[]; micState (listening, muted, error)	16-bit mono PCM in memory; typed events	Android OS calls and callbacks	Mic still on 1 s after mute: force-stop, enter MuteError, drop post-request audio (EC-04).
I1b C1 ↔ E1 Location	locationControl (start, stop)	LocationFix	Typed record	Android OS callbacks	GPS lost: keep guiding on lastLocation, say "GPS signal lost" once, send no I4 until a fix returns (EC-34).
I1c C1 ↔ E1 Driver I/O	spokenText; destination entry (before driving)	Speech, screen, notifications	Strings; LatLng	Android TTS and UI	Notification permission denied: spoken notice only (EC-24).

I2 C1 → C2	tripId, utteranceId, control (audio, cancel), audio, AudioMetadata	transcript, intent (propose, accept, reject, unknown), Constraint?, confidence (0 to 1), Error?	Binary audio frames plus JSON	Authenticated WSS (TLS)	invalid_audio 400; unsupported_constraint 422; unclear_speech 422; quota 429; provider_timeout 504. C1 asks for a repeat within 2 s and keeps the route.
I3 C2 → E2	audio, metadata, allowedConstraintTypes, allowedIntents	transcript, intent, Constraint?, confidence, ProviderUsage	Provider audio stream plus JSON	Authenticated WSS (TLS)	Timeout, quota, or invalid output becomes an I2 error. Provider must not retain audio (G-03).
I4 C1 → C3	tripId, LocationFix, destination, currentRoute?, Constraint[], tripRevision	RouteProposal? or Error	JSON	HTTPS POST	no_match 404/422; weather_unavailable 503; quota 429; timeout 504; stale_revision 409. All keep the active route.
I5 C3 → E3	origin, destination, waypoints, query?, avoidHighways, avoidTolls, alternatives	RoutePlan[], Place[], attribution	Google JSON normalized to typed records	HTTPS	No match within 600 s: no_match. Quota or timeout: preserve guidance (EC-12).
I6 C3 → E4	{location, expectedAt}[]	WeatherSample[], fetchedAt	Open-Meteo JSON normalized to typed records	HTTPS	Stale or missing data: warn and do not claim rain avoidance.
I7 C2 → C4, I8 C3 → C4	checkQuota(service); recordUsage(tripId, service, operation, units, outcome)	allow, allow_warn, deny; Unit	Typed records	In-process call	Deny returns quota to C1. Database down: queue up to 500 events, retry every 30 s, allow calls (EC-16).
I9 Developer → C4	from, to, groupBy; admin key	Counts and estimated cost per trip, day, or service	JSON	HTTPS GET	400 invalid_range; 401 unauthorized; 403 forbidden; 503 usage_store_unavailable.

3.2 Endpoint Table

Endpoint	Interface	Auth	Success response	Errors
`WSS /api/v1/trips/{tripId}/speech`	I2	Access token	Interpretation result per utterance	See I2
`POST /api/v1/trips/{tripId}/route-proposals`	I4	Access token	`{proposal: RouteProposal}`	See I4
`POST /api/v1/auth/register`	Auth	None, rate-limited per IP	accessToken, refreshToken, expiresIn (3600)	429 rate_limited; 503 auth_unavailable
`POST /api/v1/auth/refresh`	Auth	Refresh token	accessToken, expiresIn	401 invalid_refresh; 403 token_revoked
`GET /api/v1/config/client`	Config	Access token	version and client config values (no secrets)	401 unauthorized; 503 config_unavailable (app uses cache)
`GET /api/v1/usage/report?from=&to=&groupBy=`	I9	Admin key	Usage summaries	See I9

All paths use the `/api/v1/` prefix (D1 §5.4). A breaking change to any of them requires `/api/v2/`.

3.3 Message Payloads

Shared types follow D0: string = UTF-8, instant = ISO 8601 UTC, durations in seconds, distances in meters, `LatLng = {lat, lon}` in WGS84, `?` = optional.

Record	Fields
Constraint	type: add_waypoint, avoid_precipitation, avoid_highways, avoid_tolls; value: string?; maxDetourSeconds: integer?
LocationFix	position: LatLng; accuracyMeters: number; capturedAt: instant
AudioMetadata	encoding: string; sampleRateHz: integer; channels: integer; utteranceEndedAt: instant
Place	placeId: string; name: string; location: LatLng
RoutePlan	routeId: string; geometry: LatLng[]; durationSeconds; distanceMeters; steps: RouteStep[] (instruction, location, sequence)
RouteProposal	tripRevision: integer; route: RoutePlan; selectedPlace: Place?; extraSeconds: number; confirmationText: string (≤ 15 words); warnings: string[]
WeatherSample	location: LatLng; validAt: instant; precipitationProbability: number? (0 to 1); precipitationRateMmPerHour: number?

Constraint as a JSON Schema (the allow-list C2 enforces, D1 §3.1):

```
{
  "type": "object",
  "required": ["type"],
  "properties": {
    "type": {"enum": ["add_waypoint", "avoid_precipitation", "avoid_highways", "avoid_tolls"]},
    "value": {"type": "string", "minLength": 1},
    "maxDetourSeconds": {"type": "integer", "minimum": 0, "maximum": 600}
  },
  "if": {"properties": {"type": {"const": "add_waypoint"}}},
```

```
      "then": {"required": ["value"]},
      "additionalProperties": false
    }
```

I4 request and reply (D1 §5.4):

```
POST /api/v1/trips/trip-17/route-proposals
{"requestId": "req-18", "tripId": "trip-17",
 "location": {"position": {"lat": 39.1031, "lon": -84.5120}, "accuracyMeters": 8, "capturedAt": "2026-10-30T21:00:00Z"},
 "destination": {"lat": 39.9612, "lon": -82.9988}, "currentRoute": null,
 "constraints": [{"type": "add_waypoint", "value": "McDonald's", "maxDetourSeconds": 600}],
 "tripRevision": 4}

200 OK
{"proposal": {"tripRevision": 4,
  "selectedPlace": {"placeId": "place-123", "name": "McDonald's", "location": {"lat": 39.18, "lon": -84.47}},
  "extraSeconds": 420, "confirmationText": "McDonald's adds about seven minutes.", "warnings": []}}
```

3.4 Database Schema

C1 keeps its TripSession and AudioBuffer (15 s cap) in phone memory only. C2 and C3 routing state is in memory and cleared at trip end. PostgreSQL holds only the tables below (D2 §2.2, §2.3).

```
CREATE TYPE service_t AS ENUM ('ai', 'maps', 'weather');

CREATE TABLE trip_usage (            -- one row per trip; deleted after 12 months
  trip_id        TEXT PRIMARY KEY,
  ai_calls       INTEGER NOT NULL DEFAULT 0,
  maps_calls     INTEGER NOT NULL DEFAULT 0,
  weather_calls  INTEGER NOT NULL DEFAULT 0,
  est_cost_total NUMERIC NOT NULL DEFAULT 0);

CREATE TABLE usage_event (           -- one row per call attempt; deleted after 90 days
  event_id    TEXT PRIMARY KEY,      -- caller-supplied; deduplicates retries
  trip_id     TEXT NOT NULL REFERENCES trip_usage(trip_id),
  service     service_t NOT NULL,
  operation   TEXT NOT NULL,
  occurred_at TIMESTAMPTZ NOT NULL,
  units       INTEGER NOT NULL,
  est_cost    NUMERIC NOT NULL,
  outcome     TEXT NOT NULL CHECK (outcome IN ('ok', 'error', 'throttled')));
CREATE INDEX ON usage_event (trip_id);

CREATE TABLE monthly_budget (        -- kept with no limit; holds no trip data
  service service_t NOT NULL,
  month   CHAR(7) NOT NULL,          -- e.g. '2026-10'
  used    NUMERIC NOT NULL DEFAULT 0,
  state   TEXT NOT NULL CHECK (state IN ('OK', 'Warning', 'Exhausted')),
  PRIMARY KEY (service, month));

CREATE TABLE installation (
  install_id         TEXT PRIMARY KEY,
  created_at         TIMESTAMPTZ NOT NULL,
  status             TEXT NOT NULL CHECK (status IN ('active', 'revoked')),
  refresh_token_hash TEXT NOT NULL); -- raw token never stored

CREATE TABLE alert_state (           -- stops duplicate budget alerts
  service      service_t NOT NULL,
  month        CHAR(7) NOT NULL,
  level        SMALLINT NOT NULL CHECK (level IN (80, 100)),
  last_sent_at TIMESTAMPTZ NOT NULL,
  PRIMARY KEY (service, month, level));
```

The cap check is one atomic statement that refuses to pass the cap (D2 R-07), for example `UPDATE monthly_budget SET used = used + $cost WHERE service = $s AND month = $m AND used + $cost <= $cap RETURNING used;` where zero rows returned means deny.

4. Technology Stack

Layer	Choice	Justification
Mobile app	Android, Kotlin	Runs on the driver's existing phone (CON-01); Keystore for the refresh token (NFR-03); foreground service keeps mic and network alive (NFR-09); local mute path (REQ-11).
Backend	Node.js, TypeScript, Express	Integration work is I/O-bound with bounded timeouts (NFR-10); static types match the allow-list and contracts (REQ-06). MIT licensed.
Database	PostgreSQL (Render managed)	Atomic cap update (NFR-04); reliable trip and event IDs for usage records (REQ-14); retention job (NFR-08).
Speech and interpretation	OpenAI API	Hosted transcription and structured interpretation without training a model (REQ-01, REQ-06; CON-01). Accepted only with a written no-retention setting (NFR-01, G-03); fallback is on-device transcription (RK-01).
Routes, places, map	Google Maps Platform	Routes, place search, and base map (REQ-01, REQ-09, REQ-15); terms bound the design (CON-04).

Weather	Open-Meteo	Location- and time-indexed precipitation at low cost (REQ-08; CON-01).
Hosting	Render	One small web service plus database, smallest practical tier (CON-01); secrets in environment variables (NFR-03, NFR-06).
Authentication	Anonymous install tokens; 1-hour access token	No accounts needed; keys stay on the server (NFR-03).
Configuration	Versioned config file checked against a schema at startup	Tunable timeouts, thresholds, caps, retention (NFR-06).
Logging	JSON lines on Render, 30 days, redaction filter; Logcat in debug builds only	Traceable by requestId with no private data (NFR-05, NFR-01).
Transport	TLS for HTTPS and WSS	Protects audio and tokens in transit (NFR-01, NFR-03).
CI	GitHub Actions	Free for the team's GitHub repository (CON-01); runs tests with mocked providers on every push so contract changes are caught before integration (NFR-02, RK-08).

5. Known Constraints

ID	Constraint	Rules out	Forces
CON-01	Team-funded; free tiers and Maps credits; driver's own phone.	Purpose-built hardware; paid services beyond the caps; a distributed cloud setup.	One Render service and smallest database; C4 caps and 80/100% alerts (NFR-04); mocks in tests (RK-05).
CON-02	No raw audio retained; indicator and one-tap mute before any in-car demo.	Audio files, audio or transcripts in logs, debug recordings, providers that retain audio.	In-memory AudioBuffer capped at 15 s; local mute path (D02); E2 no-retention gate (G-03) with on-device fallback; trip-scoped constraints.
CON-03	ORC 4511.204: voice-operated use only while driving.	Any feature that needs a screen tap while moving, including tap-to-approve.	Spoken "okay" approval (D07); destination chosen before driving; tap mute is for passengers or a stopped driver.
CON-04	Google Maps terms: place data cached 30 days at most; no competing base map.	Swapping in a cheaper tile provider; storing provider data by default.	Google base map; any future cache expires within 30 days; if in-app guidance is not allowed, hand off to the Google Maps app (G-04).
CON-05	One short sentence per spoken reply.	Long spoken explanations.	15-word confirmation limit; EC-05 replaces long text with a place-and-minutes summary.
CON-06	Rank businesses only by stated preference and detour cost.	Paid placement; popularity ranking.	C3 filter-then-rank with a deterministic tie-break on lower extraSeconds, then shorter duration (D1 §3.2).

6. Traceability Matrix

Every REQ, NFR, and CON in Requirements Document v1.0 has one row. No row is orphaned. Test case IDs TC-01 to TC-32 are reserved for the Week 10 test plan, one per row.

Req ID	Description	Design Component (D0/D1/D2)	Status	Test Case
REQ-01	Add a stop by voice, detour ≤ 600 s	D1 §3.2 filter-then-rank; C2 add_waypoint; I4; I5	Designed	TC-01
REQ-02	Confirmation ≤ 15 words	RouteProposal.confirmationText; config confirmation.maxWords; D2 EC-05	Designed	TC-02
REQ-03	Activate only after "okay"	D2 Figure 1 AwaitingApproval; onApprovalReply; D0 D07	Designed	TC-03
REQ-04	Route active ≤ 3 s after "okay"	D2 Figure 1 Activating; activation.maxSec; D1 §3.2 proposal prepared before approval	Designed	TC-04
REQ-05	No match: keep route, prompt ≤ 3 s	C3 no_match; D2 EC-09; C1 Recovery	Designed	TC-05
REQ-06	Unparseable: keep route, prompt ≤ 2 s	D1 §3.1 allow-list validation; unclear_speech; D2 EC-07; recovery.promptSec	Designed	TC-06
REQ-07	"No" gives next candidate	D2 Figure 1 AwaitingApproval → Planning; I4	Designed	TC-07
REQ-08	Avoid rain, warn if no data	D1 §3.2 weather ranking; I6; E4; weather.* config; D2 EC-12	Designed	TC-08
REQ-09	Avoid highways and tolls	D1 §3.2 avoidance flags; I5 avoidHighways/avoidTolls	Designed	TC-09
REQ-10	Mic state always shown	D2 TripSession.micState; I1a; foreground notification (§2.3.2)	Designed	TC-10
REQ-11	Mute ≤ 1 s by tap or voice	D2 mute(); I1a; D0 D02; Figure 1 Muted	Designed	TC-11
REQ-12	Force-stop ≤ 2 s, 0 s post-mute audio	D2 MuteError; AudioBuffer; EC-04	Designed	TC-12
REQ-13	Unmute by tap only	D2 unmute(); Figure 1 Muted → Listening	Designed	TC-13
REQ-14	Per-trip usage counts and report	D2 §2.2 UsageEvent, TripUsage, recordUsage; I7, I8, I9	Designed	TC-14
REQ-15	Turn-by-turn spoken guidance	D0 C1; I1b, I1c; I5 RouteStep	Designed	TC-15
NFR-01	0 raw audio files; no-retention provider	D2 AudioBuffer; logging redaction; D0 D03; I3 provider gate	Designed	TC-16
NFR-02	Keep route on any failure	D2 Figure 1 Recovery; EC-01 to EC-34; D0 D08	Designed	TC-17
NFR-03	No keys in app; Keystore; 1 h tokens	D2 §2.3.1 Installation, /api/v1/auth/register, /api/v1/auth/refresh	Designed	TC-18
NFR-04	Alert at 80%, deny at 100%, no overrun	D2 §2.2 MonthlyBudget, checkQuota; R-07; EC-13, EC-14	Designed	TC-19

NFR-05	JSON logs with no private data	D2 §2.3.4 logger, requestId, redaction filter; EC-29	Designed	TC-20
NFR-06	All tunables in validated config	D2 §2.3.3 config schema; GET /api/v1/config/client; EC-25 to EC-28	Designed	TC-21
NFR-07	≤ 15-word replies; no taps while moving	D0 D09; confirmation.maxWords; I1c; notifyDriver	Designed	TC-22
NFR-08	90-day / 12-month retention	D2 §2.2 retention job; retention.* config	Designed	TC-23
NFR-09	Mic and network work in background	D2 §2.3.2 foreground notification; I1a	Designed	TC-24
NFR-10	p95 latency for I2 and I4 under load	D1 §3.1 and §3.2 complexity and timeouts; timeout.interpretSec, planSec	Planned	TC-25
NFR-11	≥ 8 of 10 phrases correct, with accents	D1 §3.1 allow-list validation; task T10; milestone M-18	Planned	TC-26
CON-01	Team-funded, existing phone	D0 §6 project constraints; D01; C4 caps	Designed	TC-27
CON-02	No raw audio; indicator and mute	D0 D02, D03; D2 AudioBuffer, mute()	Designed	TC-28
CON-03	Voice-only use while moving	D0 D07; D2 I1c	Designed	TC-29
CON-04	Google Maps caching and base-map terms	D0 D05; D1 §6.5	Designed	TC-30
CON-05	One short sentence per reply	D0 D09; D2 EC-05	Designed	TC-31
CON-06	Rank by preference and detour only	D0 D10; D1 §3.2 filter-then-rank	Designed	TC-32

7. Advisor Review Record

Fill this in after the review meeting with Dr. Aurisano, then add a v1.1 row to both change logs listing each edit the review produced.

Item	Record
Date drafts sent to advisor	
Date of review meeting	
Attendees	
Documents reviewed	Requirements Document v1.0; Technical Specification v1.0 (including the traceability matrix)
What was discussed	
Changes it produced	